



Sri Chaitanya IIT Academy., India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: **Sr.Super60_NUCLEUS&STERLING_BT**

JEE-MAIN

Date: 04-11-2023

Time: **09.00Am to 12.00Pm**

RPTM-13

Max. Marks: 300

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black Ball Point Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of 90 questions. The maximum marks are **300**.
5. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and subject having **two sections**.

(I) Section –I contains 20 **multiple choice** questions with only one correct option.

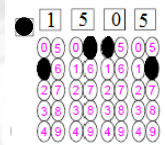
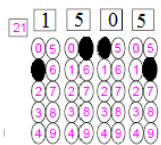
Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

(II) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions only, if more than 5 questions attempted, First 5 attempted questions will be considered.

- The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

To cancel any attempted question bubble on the question number box.

For example: To cancel attempted question 21. Bubble on 21 as shown below



Question Answered for Marking

Question Cancelled for Marking

Marking scheme: +4 for correct answer, 0 if **not attempt** and -1 in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold of make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number:

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Candidate's Signature: _____

Invigilator's Signature: _____



04-11-23_Sr.Super60_NUCLEUS & STERLING_BT_Jee-Main_RPTM-13

Q Paper Prepared as per Jee Main 2024 New Syllabus**PHYSICS**

: Electrostatics, Gauss law & Gravitation: Coulomb's law; Electric field and potential; Electrical potential energy of a system of point charges and of electrical dipoles in a uniform electrostatic field; Electric field lines; Flux of electric field; Gauss's law and its application in simple cases, such as, to find field due to infinitely long straight wire, uniformly charged infinite plane sheet and uniformly charged thin spherical shell. Dielectrics and electric polarization Law of gravitation; Gravitational potential and field; Acceleration due to gravity; Kepler's law, Geostationary orbits, Motion of planets and satellites in circular orbits; Escape velocity.

RPTM-13(15Q-RPTM.13 SYLLABUS+15Q CUMULATIVE SYLLABUS)

CHEMISTRY

: 1.f-block elements: General characteristics ; electronic configurations lanthanoids and actinoids ; lanthanoid and actinoid contraction ; oxidation states ; properties ; applications

RPTM-13(15Q-RPTM.13 SYLLABUS+15Q CUMULATIVE SYLLABUS)

MATHEMATICS : Total Conic Section (Parabola, Ellipse & Hyperbola)

RPTM-13(15Q-RPTM.13 SYLLABUS+15Q CUMULATIVE SYLLABUS)

Sri Chaitanya Educational Institutions
INDIA





PHYSICS

Max Marks: 100

(SINGLE CORRECT ANSWER TYPE)

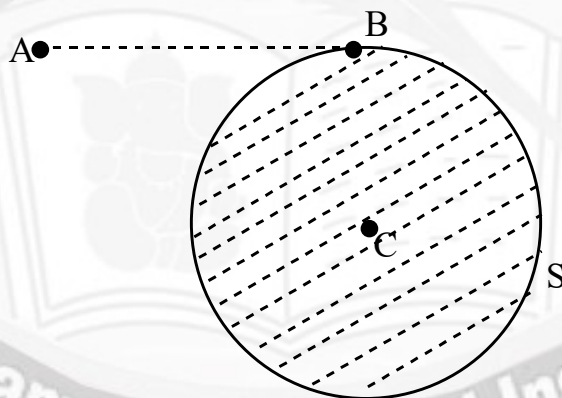
This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

1. A thin infinite sheet charge and an infinite line charge of respective charge densities $+\sigma$ and $+\lambda$ are placed parallel at 8 m distance from each other. Points 'P' and 'Q' are at $\frac{3}{\pi}$ m and $\frac{4}{\pi}$ m perpendicular distances from line charge towards sheet charge, respectively.

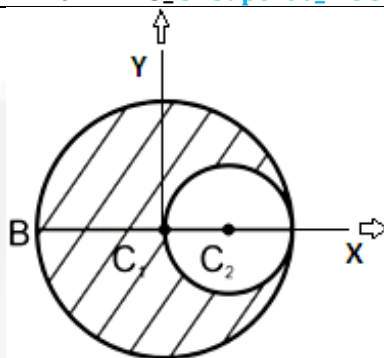
' E_P ' and ' E_Q ' are the magnitudes of resultant electric field intensities at point 'P' and 'Q' respectively. If $2|\sigma| = |\lambda|$, the ratio of $\frac{E_P}{E_Q}$ is _____

- 1) 2:3 2) 8:9 3) 3:2 4) 9:8
2. S is a solid neutral conducting sphere. A point charge $q = 1 \times 10^{-6} \text{ C}$ is placed at point A. C is the centre of sphere and AB is a tangent $BC = 3 \text{ m}$ and $AB = 4 \text{ m}$. The electric potential at B due to induced charge on the sphere is



- 1) 1.2kV 2) -1.2kV 3) -0.45kV 4) 0.45kV
3. A positively charged sphere of radius r_0 carries a volume charge density ρ . A spherical cavity of radius $r_0 / 2$ is then hollowed out and this portion is left empty as cavity as shown. C_1 is the centre of the sphere and C_2 is that of the cavity the direction and magnitude of the electric field at point B is





- 1) $\frac{17\rho r_0}{54\epsilon_0}(-\hat{i})$ 2) $\frac{\rho r_0}{6\epsilon_0}(-\hat{i})$ 3) $\frac{17\rho r_0}{54\epsilon_0}(\hat{i})$ 4) $\frac{\rho r_0}{6\epsilon_0}(\hat{i})$

4. Two point dipoles of dipole moments $p\hat{k}$ & $\frac{p}{2}\hat{k}$ are located at $(0,0,0)$ & $(1m,0,2m)$ respectively. The resultant electric field due to the two dipoles at the point $(1m,0,0)$ is:

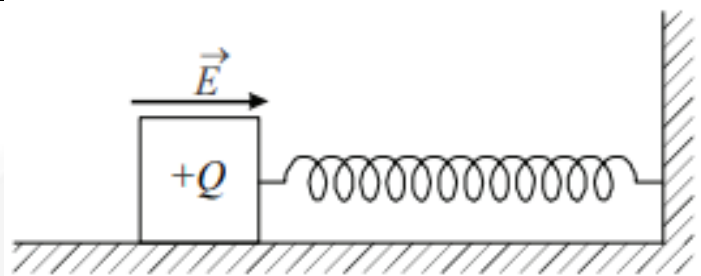
- 1) $\frac{9p}{32\pi\epsilon_0}\hat{k}$ 2) $\frac{7p}{32\pi\epsilon_0}(-\hat{k})$ 3) $\frac{7p}{32\pi\epsilon_0}\hat{k}$ 4) 0

5. Consider a system of three charges $\frac{q}{3}, \frac{q}{3}$ and $-\frac{2q}{3}$ placed at points A, B and C respectively which lie on a circle with O to be the center with radius R and angle $CAB = 60^\circ$. If AB is the diameter of the circle, choose the correct statement

- 1) The electric field at point O is $\frac{q}{8\pi\epsilon_0 R^2}$ directed along the negative x-axis.
 2) The potential energy of the system is zero.
 3) The magnitude of the electrostatic force between the charges at C and B is $\frac{q^2}{54\pi\epsilon_0 R^2}$
 4) The potential at point O is $\frac{q}{12\pi\epsilon_0 R}$

6. A wooden block performs SHM on a frictionless surface with frequency ν_0 . The block carries a charge $+Q$ on its surface. If now a uniform electric field $\vec{E}$ is switched-on as shown, then the SHM of the block will be :





- 1) Of the same frequency and with shifted mean position.
 - 2) Of the same frequency and with the same mean position.
 - 3) Of changed frequency and with shifted mean position.
 - 4) Of changed frequency and with the same mean position.
7. A Satellite is revolving round the Earth with orbital speed V_0 . If it stops suddenly, the speed with which it will strike the surface of the Earth would be (V_e = escape velocity of a particle on Earth's surface)
- 1) $\frac{V_e^2}{V_0}$
 - 2) $\sqrt{V_e^2 - V_0^2}$
 - 3) $\sqrt{V_e^2 - 2V_0^2}$
 - 4) $\frac{V_0}{2}$
8. A cord of length 64 m is used to connect a 100 kg astronaut to spaceship whose mass is much larger than that of the astronaut. Estimate the value of the tension in the cord. Assume that the spaceship is orbiting near earth surface. Assume that the spaceship and the astronaut fall on a straight line from the earth centre. The radius of the earth is 6400 km. ($g = 10 \text{ m/s}^2$)
- 1) $T = 5 \times 10^{-2} \text{ N}$
 - 2) $T = 2 \times 10^{-2} \text{ N}$
 - 3) $T = 3 \times 10^{-2} \text{ N}$
 - 4) $T = 4 \times 10^{-2} \text{ N}$
9. The gravitational potential energy of interactions of a system of six identical particles, each of mass m placed at the vertices of a regular hexagon of side 'a' is [PE = 0 at infinite separation]
- 1) $-\frac{Gm^2}{a} \left(1 + \frac{1}{\sqrt{3}} + \frac{1}{4} \right)$
 - 2) $-\frac{3Gm^2}{a} \left(1 + \frac{1}{\sqrt{3}} + \frac{1}{4} \right)$
 - 3) $-\frac{6Gm^2}{a} \left(1 + \frac{1}{\sqrt{3}} + \frac{1}{4} \right)$
 - 4) $-\frac{2Gm^2}{a} \left(1 + \frac{1}{\sqrt{3}} + \frac{1}{4} \right)$





10. An object of mass m lying at rest at depth of ' $R/2$ ' from the Earth's surface, is to made an Earth's satellite at a height of ' $R/2$ ' from the Earth's surface. The work required to do this is... (M-mass of the Earth, R-radius of the Earth and neglect the rotation of earth.)
- 1) $\frac{GMm}{R}$ 2) $\frac{25GMm}{24R}$ 3) $\frac{17GMm}{24R}$ 4) $\frac{15GMm}{2R}$
11. Statement I: If there is no external torque on a body about its center of mass, then the velocity of the center of mass remains constant.
Statement II: The linear momentum of an isolated system ($F_{ext} = 0$) remains constant.
- 1) Both Statement I and Statement II are incorrect
2) Both Statement I and Statement II are correct
3) Statement I is True, Statement II is False
4) Statement I is False, Statement II is True
12. Statement I: Two cylinders, one hollow (metal) and the other solid (wood) with the same mass and identical dimensions are simultaneously allowed to roll without slipping down an inclined plane from the same height. The hollow cylinder will reach the bottom of the inclined plane first.
Statement II: By the principle of conservation of energy, the total kinetic energies of both the cylinders are identical when they reach the bottom of the incline.
- 1) Statement I is True, Statement II is True; Statement II is a correct explanation for Statement I
2) Statement I is True, Statement II is True; Statement II is NOT a correct explanation for Statement I
3) Statement I is True, Statement II is False
4) Statement I is False, Statement II is True





13. **Assertion (A) :** In a given situation of arrangement of charges, an additional charge is placed outside the Gaussian surface. In this situation, in the Gauss theorem $\oint \vec{E} \cdot d\vec{s} = \frac{q_{in}}{\epsilon_0}$, q_{in} remains unchanged whereas electric field $\vec{E}$ is changed.
- Reason (R) :** Electric field $\vec{E}$ at any point on the Gaussian surface is due to inside charge only.
- 1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - 2) Both (A) and (R) are true and (R) is NOT the correct explanation of (A).
 - 3) (A) is true but (R) is false.
 - 4) (A) is false but (R) is true.
14. Statement I: The stream of water flowing at high speed from a garden hose pipe tends to spread like a fountain when held vertically up, but tends to narrow down when held vertically down.
- Statement II: In any steady flow of an incompressible fluid, the volume flow rate of the fluid remains constant.
- 1) Statement I is True, Statement II is True; Statement II is a correct explanation for Statement I
 - 2) Statement I is True, Statement II is True; Statement II is NOT a correct explanation for Statement I
 - 3) Statement I is True, Statement II is False
 - 4) Statement I is False, Statement II is True
15. **Assertion (A) :** If dQ and dW represent the heat supplied to the system and the work done on the system respectively. Then according to the first law of thermodynamics $dQ = dU - dW$
- Reason (R) :** First law of thermodynamics is based on law of conservation of energy.
- 1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - 2) Both (A) and (R) are true and (R) is NOT the correct explanation of (A).
 - 3) (A) is true but (R) is false.
 - 4) (A) is false but (R) is true.

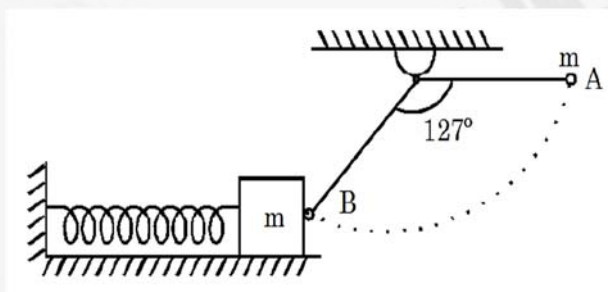




16. The horizontal range of a projectile is R and the maximum height attained by it is H . A strong wind now begins to blow in the direction of the motion of the projectile, giving it a constant horizontal acceleration $=g/2$. Under the same conditions of projection, the horizontal range of the projectile will now be

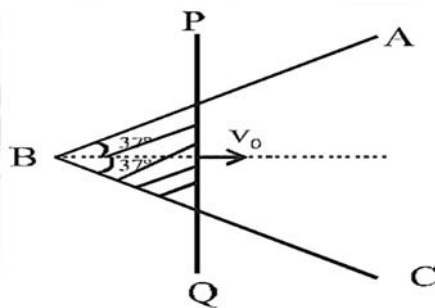
1) $R + \frac{H}{2}$ 2) $R + H$ 3) $R + \frac{3H}{2}$ 4) $R + 2H$

17. A pendulum of length ℓ and mass m is released from horizontal position A as shown and strikes elastically a block of same mass connected to a spring of constant K placed on a horizontal smooth surface. Find the maximum deformation in the spring.



1) $\frac{8}{5}\sqrt{\frac{2mg\ell}{5K}}$ 2) $\frac{8}{5}\sqrt{\frac{mg\ell}{5K}}$ 3) $\frac{4}{5}\sqrt{\frac{2mg\ell}{5K}}$ 4) $\frac{4}{5}\sqrt{\frac{2mg\ell}{K}}$

18. ABC is a smooth horizontal wire frame. PQ is a wire of mass m , which can slide on frame. There is a film of a liquid of surface tension T covering the triangular part. Wire PQ is given a velocity v_0 when it is at a distance x_0 from B. Find v_0 if wire PQ is just able to reach upto a distance $2x_0$ from B.



1) $x_0\sqrt{\frac{T}{m}}$ 2) $2x_0\sqrt{\frac{T}{m}}$ 3) $3x_0\sqrt{\frac{T}{m}}$ 4) $4x_0\sqrt{\frac{T}{m}}$



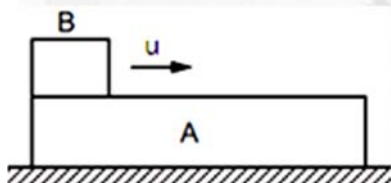


19. Four charges Q_1, Q_2, Q_3 and Q_4 of same magnitude are fixed along the x axis at $x = -2a, -a, +a$ and $+2a$, respectively. A positive charge q is placed on the positive y axis at a distance $b > 0$. Four options of the signs of these charges are given in List I. The direction of the forces on the charge q is given in List II. Match List I with List II and select the correct answer using the code given below the lists.

List I	List II
P) Q_1, Q_2, Q_3 and Q_4 all positive	1) +x
Q) Q_1 and Q_2 are positive, Q_3 and Q_4 are negative	2) -x
R) Q_1 and Q_4 are positive, Q_2 and Q_3 are negative	3) +y
S) Q_1 and Q_3 are positive, Q_2 and Q_4 are negative	4) -y

- 1) P-3, Q-1, R-4, S-2 2) P-4, Q-2, R-3, S-1
 3) P-3, Q-1, R-2, S-4 4) P-4, Q-2, R-1, S-3

20.



A long block A is at rest on a smooth horizontal surface. A small block B, whose mass is half of A, is placed on A at one end and projected along A with some velocity u . The coefficient of friction between the blocks is μ , then choose the incorrect option

- 1) The blocks will reach a final common velocity $\frac{u}{3}$
 2) The work done against friction is two-third of the initial kinetic energy of B.
 3) Before the blocks reach a common velocity the magnitude of acceleration of A relative to B is $\frac{2}{3}\mu g$
 4) Before the blocks reach a common velocity the magnitude of acceleration of A relative to B is $\frac{3}{2}\mu g$





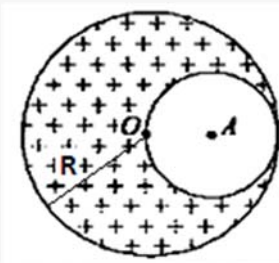
(NUMERICAL VALUE TYPE)

Section-II contains 10 Numerical Value Type questions. Attempt any 5 questions only. First 5 attempted questions will be considered if more than 5 questions attempted. The Answer should be within 0 to 9999. If the Answer is in Decimal then round off to the nearest Integer value (Example i.e. If answer is above 10 and less than 10.5 round off is 10 and If answer is from 10.5 and less than 11 round off is 11).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. A solid sphere having uniform charge density ρ and radius R is shown in figure. A spherical

cavity of radius $\frac{R}{2}$ is made in it. The potential at point O is $\frac{5R^2\rho}{n\epsilon_0}$, then n is ____



22. A solid conducting sphere having a charge Q is surrounded by an uncharged concentric conducting hollow spherical shell. Let the potential difference between the surface of the solid sphere and that of the outer surface of the hollow shell be V . If the shell is now given a charge of $-3Q$, then new potential difference between the same two surfaces is nV . Then n is ____

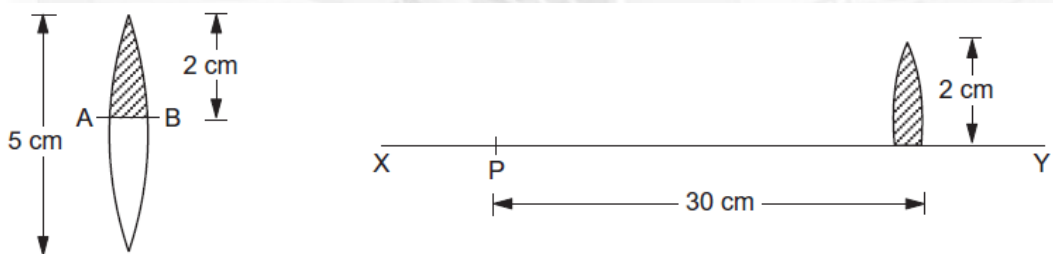
23. The electric field intensity at the centre of a uniformly charged hemispherical shell is E_0 . Now, two portions of the hemisphere are cut from either side and remaining portion is shown in figure. If $\alpha = \beta = \frac{\pi}{3}$ then electric field intensity at centre due to remaining portion is $\frac{E_0}{n}$, then n is ____



24. Three concentric spherical metallic shells X, Y and Z of radius a , b and c respectively [$a < b < c$] have surface charge densities σ , $-\sigma$ and σ , respectively. The shells X and Z are at same potential. If the radii of X & Y are 3 cm and 4 cm, respectively. The radius of shell Z is ____ cm.





25. A smooth tunnel is dug along the radius of the earth (R) that ends at the centre of earth. A ball is released from the surface of earth along the tunnel. If the coefficient of restitution is 0.2 between the end of tunnel and ball then the total distance travelled by the ball before second collision at the centre of earth is $\frac{xR}{5}$. Then x is _____
26. A beam of light consisting of two wavelengths 7000Å and 5500Å is used to obtain interference pattern in Young's double slit experiment. The distance between the slits is 2.5 mm and the distance between the place of slits and the screen is 150 cm. The least distance from the central fringe, where the bright fringes due to both the wavelengths coincide, is $n \times 10^{-5}\text{ m}$. The value of n is _____.
27. The area of cross-section of a railway track is 0.01m^2 . The temperature variation is 10^0C . Coefficient of linear expansion of material of track is $10^{-5}/^\circ\text{C}$. The energy stored per meter in the track is _____ J/m.
(Young's modulus of material of track is 10^{11}Nm^{-2})
28. 
29. The amplitude of wave disturbance propagating in the positive x -direction is given by $y = \frac{1}{1+x^2}$ at time $t=0$ and $y = \frac{1}{1+(x-2)^2}$ at $t=1\text{s}$, where x and y are in metres. The shape of wave does not change during the propagation. The velocity of the wave will be _____ m/s.
30. A system consists of two types of gas molecules A and B having same number density $2 \times 10^{25}/\text{m}^3$. The diameter of A and B are 10Å and 5Å respectively. They suffer collision at room temperature. The ratio of average distance covered by the molecule A to that of B between two successive collision is _____ $\times 10^{-2}$



**CHEMISTRY****Max Marks: 100****(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

31. Which of the following actinoids have one electron in 6d orbital?
 I) U ($Z = 92$) II) Np ($Z = 93$) III) Pu ($Z = 94$) IV) Am ($Z = 95$)
 1) I and III only 2) I and IV only 3) I and II only 4) only III
32. Assertion(A) : Zr(160pm) and Hf(159pm) have almost identical radii
 Reason(R) : Zr and Hf lie in the same group of the periodic table
 1) Both A and R are true, and R is the correct explanation of the A
 2) Both A and R are true, but R is not the correct explanation of A
 3) A is not true but R is true
 4) Both A and R are false
33. Arrange the following in decreasing order of their boiling points.
 (i) n-butane (ii) 2-methylbutane (iii) n-pentane (iv) 2,2-dimethylpropane
 1) i > ii > iii > iv 2) ii > iii > iv > i 3) iv > iii > ii > i 4) iii > ii > iv > i
34. Larger number of oxidation states are exhibited by the actinoids than those by the lanthanoids, the main reason being
 1) Lesser energy difference between 5f and 6d than between 4f and 5d orbitals
 2) More energy difference between 5f and 6d than between 4f and 5d orbitals
 3) More reactive nature of the actinoids than the lanthanoids
 4) 4f orbitals more diffused than the 5f orbitals
35. Statement I: In general, actinoids form relatively less stable complexes as compared to lanthanoids
 Statement II: Actinoid contraction is greater from element to element than lanthanoid contraction
 1) Both Statement I and Statement II are true, and Statement II is the correct explanation of the Statement I



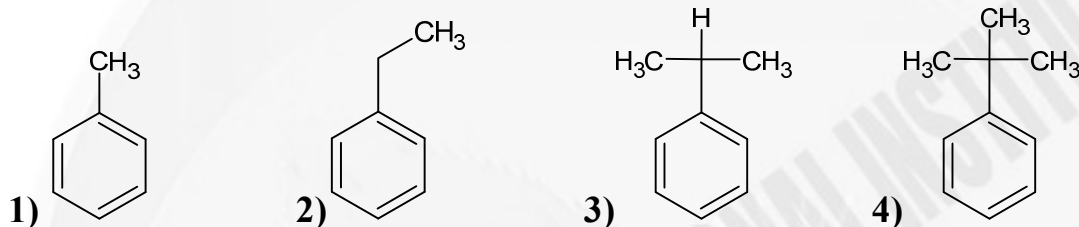


2) Both Statement I and Statement II are true, but Statement II is not the correct explanation of Statement I

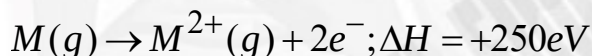
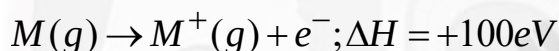
3) Statement I is not true but Statement II is true

4) Both I and II are false

36. In which of the following case, the rate of Electrophilic Aromatic Substitution will be faster.



37. Consider the following ionization steps:



Select correct statement(s)

I) $I.E._1$ of $M(g)$ is 100eV

II) $I.E._1$ of $M^+(g)$ is 150eV

III) $I.E._2$ of $M(g)$ is 250eV

IV) $I.E._2$ of $M(g)$ is 150eV

1) I, III

2) I, II, III

3) I, II, IV

4) I, II, III, IV

38. Which of the following is not correctly match between given species and type of overlapping?

1) XeO_3 : Three ($d\pi - p\pi$) bonds

2) H_2SO_4 : Two ($d\pi - p\pi$) bonds

3) SO_3 : Three ($d\pi - p\pi$) bonds

4) HClO_4 : Three ($d\pi - p\pi$) bonds

39. A diatomic molecule has a dipole moment of 1.2D. If the bond length is $1.0 \times 10^{-8}\text{cm}$, what fraction of charge does exist on each atom?

1) 0.1

2) 0.2

3) 0.25

4) 0.3





40. Which of the following statements is incorrect?
- 1) Pr, Nd, Tb and Dy exhibit +4 state in oxides (MO_2) only
 - 2) Ce(IV) acts as a good reducing agent
 - 3) In lanthanoids, the decrease in atomic radii (derived from the structures of metals) is not quite regular as it is regular in M^{3+} ions
 - 4) Eu^{+2} , Yb^{+2} acts as reducing agents
41. Arrange the following cyano complexes in increasing order of magnetic moment.
- 1) $[\text{Fe}(\text{CN})_6]^{4-} < [\text{Fe}(\text{CN})_6]^{3-} < [\text{Mn}(\text{CN})_6]^{3-} < [\text{Cr}(\text{CN})_6]^{3-}$
 - 2) $[\text{Fe}(\text{CN})_6]^{3-} < [\text{Cr}(\text{CN})_6]^{3-} < [\text{Mn}(\text{CN})_6]^{3-} < [\text{Fe}(\text{CN})_6]^{4-}$
 - 3) $[\text{Mn}(\text{CN})_6]^{3-} < [\text{Fe}(\text{CN})_6]^{3-} < [\text{Fe}(\text{CN})_6]^{4-} < [\text{Cr}(\text{CN})_6]^{3-}$
 - 4) $[\text{Fe}(\text{CN})_6]^{3-} < [\text{Cr}(\text{CN})_6]^{3-} < [\text{Fe}(\text{CN})_6]^{4-} < [\text{Mn}(\text{CN})_6]^{3-}$
42. Which of the following species can act as reducing agent?
- 1) $[\text{Co}(\text{CO})_4]^-$
 - 2) $[\text{Mn}(\text{CO})_6]$
 - 3) $[\text{Mn}(\text{CO})_5]$
 - 4) $[\text{Cr}(\text{CO})_6]$
43. **Statement-1:** 5f orbitals resemble the 4f orbitals in their angular part of the wave-function.
Statement-2: 5f orbitals are not as buried as 4f orbitals and hence 5f electrons can participate in bonding to a far greater extent.
- 1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 - 2) Statement-1 is True, Statement-2 is True, Statement-2 is NOT a correct explanation for Statement-1.
 - 3) Statement-1 is True, Statement-2 is False.
 - 4) Statement-1 is False, Statement-2 is True.





44. The radius of La^{3+} is $1.06\text{\AA}$, which of the following given values will be closest to the radius of Yb^{+3} ?
- (Given that: Atomic numbers: La = 57, Yb = 70)
- 1) $1.6\text{\AA}$ 2) $1.4\text{\AA}$ 3) $1.06\text{\AA}$ 4) $0.86\text{\AA}$
45. Periodic position of the lanthanoids(Ce to Lu) is(modern periodic table):
- 1) Group-3 ; Period – 7th 2) Group-2; Period – 6th
 3) Group-3; Period – 6th 4) Group-4; Period-4th
46. $(X) + \text{K}_2\text{CO}_3 + \text{Air} \xrightarrow{\text{heat}} (Y)$
 $(Y) + \text{Cl}_2 \rightarrow (Z)$ Pink
 Which of the following is correct?
- 1) $X = \text{black}, \text{MnO}_2, Y = \text{Blue}, \text{K}_2\text{CrO}_4, Z = \text{KMnO}_4$
 2) $X = \text{green}, \text{Cr}_2\text{O}_3, Y = \text{Yellow}, \text{K}_2\text{CrO}_4, Z = \text{K}_2\text{Cr}_2\text{O}_7$
 3) $X = \text{black}, \text{MnO}_2, Y = \text{green}, \text{K}_2\text{MnO}_4, Z = \text{KMnO}_4$
 4) $X = \text{black}, \text{Bi}_2\text{O}_3, Y = \text{colourless KBiO}_2, Z = \text{KBiO}_3$
47. Statement-1: Lanthanum ($Z=57$) exhibits +4 oxidation state.
 Statement-2: After the loss of 6s and 5d electrons, lanthanum achieves inert gas like configuration.
- 1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 2) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1
 3) Statement-I is True, Statement-2 is False.
 4) Statement-I is False, Statement-2 is True.
48. The actinoids showing +7 oxidation state are:
- 1) U, Np 2) Pu, Am 3) Np, Pu 4) Am, Cm
49. If order of the size of lanthanide ions are in order $P^{3+} > Q^{3+} > R^{3+} > S^{3+}$ then P,Q,R and S are respectively:
- 1) Ce, Pm, Eu, Lu 2) Lu, Eu, Pm, Ce 3) Pm, Ce, Lu, Eu 4) Lu, Ce, Pm, Eu





50. Select CORRECT order of basic strength.

- 1) $Ce(OH)_3 > Lu(OH)_3 > Gd(OH)_3$ 2) $Lu(OH)_3 > Gd(OH)_3 > Ce(OH)_3$
 3) $Ce(OH)_3 > Gd(OH)_3 > Lu(OH)_3$ 4) $Gd(OH)_3 > Lu(OH)_3 > Ce(OH)_3$

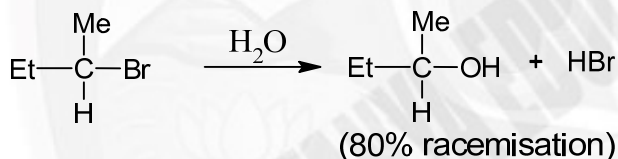
(NUMERICAL VALUE TYPE)

Section-II contains 10 Numerical Value Type questions. Attempt any 5 questions only. First 5 attempted questions will be considered if more than 5 questions attempted. The Answer should be within 0 to 9999. If the Answer is in Decimal then round off to the nearest Integer value (Example i.e. If answer is above 10 and less than 10.5 round off is 10 and If answer is from 10.5 and less than 11 round off is 11).

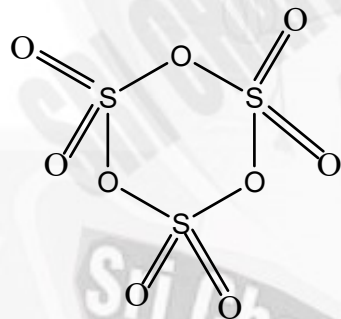
Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

51. Calculate value of $\frac{X+Y+Z}{10}$, here X is O-N-O bond angle in NO_3^- , Y is O-N-O bond angle in NO_2^+ and Z is F-Xe-F adjacent bond angle in XeF_4 .

52. The percentage of backside attack in Solvolysis reaction given below is X%. What is value of X



53. Consider the following molecule



Calculate value of $p \div q$, here p and q are total number of $d\pi - p\pi$ bonds and total number of sp^3 hybridised atoms respectively in given molecule.





54.

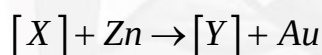
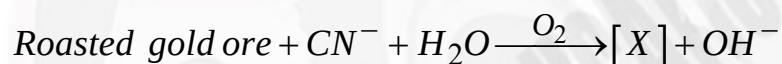
Actinoid ion	Number of unpaired electrons
$U^{+5} (Z = 92)$	A
$Pu^{+6} (Z = 94)$	B

Then (A+B) is _____

55. The total no. of unpaired electrons in Gd(z=64) are?

56. Per monomeric unit, how much is the increase in molar mass(in gram) when cellulose is treated with acetic anhydride in excess?

57. In the process of extraction of gold,

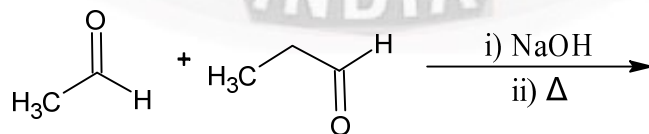


What is the sum of oxidation number of central metals in compound X and compound Y.

58. Find the total number of feasible products (including stereoisomers) in the following reaction. (No carbocation rearrangement is observed)

59. 0.93g of aniline is Kjeldahlised and the liberated ammonia gas passed into 200ml of 0.05M H_2SO_4 . How much volume of 0.1M NaOH required to neutralise unreacted H_2SO_4 ?

60. Find the total number (including stereo isomers) of aldol (Both self and crossed) condensation product(s) formed in the given reaction is:





MATHEMATICS

Max Marks: 100

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

61. Let M denote the matrix $\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$, where $i^2 = -1$, and I denote the identity matrix $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. Then the matrix $I + M + M^2 + M^3 + M^4 + \dots + M^{2023}$ is equal to:
- 1) $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ 2) $\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$ 3) $\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$ 4) $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$
62. Let $a, b \in \mathbb{R}$. If the mirror image of the point $P(a, 6, 9)$ with respect to the line $\frac{x-3}{7} = \frac{y-2}{5} = \frac{z-1}{-9}$ is $(20, b, -a-9)$, then $|a+b|$ is equal to
- 1) 88 2) 86 3) 84 4) 90
63. Let $\vec{a} = \hat{i} + \alpha\hat{j} + 3\hat{k}$ and $\vec{b} = 3\hat{i} - \alpha\hat{j} + \hat{k}$. If the area of the parallelogram whose adjacent sides are represented by the vectors $\vec{a}$ and $\vec{b}$ is $8\sqrt{3}$ square units, then $\vec{a} \cdot \vec{b}$ is equal to
- 1) 1 2) 2 3) 4 4) 8
64. Let $f(x)$ and $g(x)$ be continuous, positive functions such that $f(-x) = g(x) - 1$, $f(x) = \frac{g(x)}{g(-x)}$ and $\int_{-20}^{20} f(x) dx = 2024$, then $\int_{-20}^{20} \frac{f(x)}{g(x)} dx$ is
- 1) 1012 2) 506 3) 2024 4) 2012
65. The number of points where $f(x) = |x + [x]| - 3[2x] + 4[3x]$ is discontinuous in $[-1, 1]$ is [Note: $[k]$ denotes greatest integer less than or equal to k .]
- 1) 9 2) 8 3) 7 4) 6
66. Let f be a differentiable function such that $\lim_{x \rightarrow 1} \frac{f(1+x^3-x) - f(x)}{\sin(x-1)} = \lim_{x \rightarrow 0} \frac{f(1-x) - f(1)}{x} + 10$, then $f'(1)$ is equal to:
- 1) 5 2) 4 3) 2 4) 1





67. The value of the integral

$$\int \frac{\sin \theta \cdot \sin 2\theta (\sin^6 \theta + \sin^4 \theta + \sin^2 \theta) \sqrt{2 \sin^4 \theta + 3 \sin^2 \theta + 6}}{1 - \cos 2\theta} d\theta \text{ is:}$$

(where c is a constant of integration)

1) $\frac{1}{18} \left[11 - 18 \sin^2 \theta + 9 \sin^4 \theta - 2 \sin^6 \theta \right]^{\frac{3}{2}} + c$

2) $\frac{1}{18} \left[9 - 2 \cos^6 \theta - 3 \cos^4 \theta - 6 \cos^2 \theta \right]^{\frac{3}{2}} + c$

3) $\frac{1}{18} \left[9 - 2 \sin^6 \theta - 3 \sin^4 \theta - 6 \sin^2 \theta \right]^{\frac{3}{2}} + c$

4) $\frac{1}{18} \left[11 - 18 \cos^2 \theta + 9 \cos^4 \theta - 2 \cos^6 \theta \right]^{\frac{3}{2}} + c$

68. The triangle of maximum area the can be inscribed in a given circle of radius 'r' is :

1) An isosceles triangle with base equal to 2r.

2) An equilateral triangle of height $\frac{2r}{3}$.

3) An equilateral triangle having each of its side of length $\sqrt{3}r$.

4) A right angle triangle having two of its sides of length 2r and r.

69. The solution of differential equation $\frac{dy}{dx} = \frac{x^2 + y^2 + 1}{2xy}$ satisfying $y(1) = 0$ is given by:

1) a circle 2) $y^2 = x^2 + x - 2$ 3) a hyperbola 4) an ellipse

70. The system of linear equations $x + \mu y - z = 0, \mu x - y - z = 0, x + y - \mu z = 0$, has a non-trivial solution for:

1) Exactly three values of μ

2) Infinitely many values of μ

3) Exactly one value of μ

4) Exactly two values of μ





71. Choose the correct statement about two circles whose equations are given below:
 $x^2 + y^2 - 10x - 10y + 41 = 0$
 $x^2 + y^2 - 22x - 10y + 137 = 0$
 1) Circles have same centre 2) Circles have no meeting point
 3) Circles have only one meeting point 4) Circles have two meeting points
72. Let ABCD be a parallelogram, the equations of whose diagonals are $AC: x + 2y - 3 = 0$ and $BD: 2x + y - 3 = 0$. If the length of the diagonal $AC = 4$ units and the area of the parallelogram $[ABCD] = 8$ square units. The length of diagonal BD, is:
 1) $\frac{20}{3}$ 2) 5 3) $\frac{10}{3}$ 4) 2
73. The locus of the point of trisection of all the double ordinates of the parabola $y^2 = lx$ is a parabola whose length of latus rectum is:
 1) $\frac{l}{9}$ 2) $\frac{2l}{9}$ 3) $\frac{4l}{9}$ 4) $\frac{l}{39}$
74. If the point of intersections of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and the circle $x^2 + y^2 = 4b, b > 4$ lie on the curve $y^2 = 3x^2$, then b is equal to:
 1) 12 2) 5 3) 6 4) 10
75. The graph of the equation $x + y = x^3 + y^3$ is the union of:
 1) line and an ellipse 2) line and a parabola
 3) line and hyperbola 4) line and a point
76. Let $S(5,12)$ and $S'(-12,5)$ are the foci of an ellipse passing through the origin. The eccentricity of ellipse equals:
 1) $\frac{1}{2}$ 2) $\frac{1}{\sqrt{3}}$ 3) $\frac{1}{\sqrt{2}}$ 4) $\frac{2}{3}$





77. The y-axis is the directrix of the ellipse with eccentricity $e = \frac{1}{2}$ and the corresponding focus is at $(3,0)$, equation to its auxiliary circle is:

1) $x^2 + y^2 - 8x + 12 = 0$ 2) $x^2 + y^2 - 8x - 12 = 0$

3) $x^2 + y^2 - 8x + 9 = 0$ 4) $x^2 + y^2 = 4$

78. Statement I: The latus rectum is the shortest focal chord of a parabola.

Statement II: Length of a focal chord of the parabola $y^2 = 4ax$ is $a\left(t + \frac{1}{t}\right)^2$, at point 't',

which is minimum when $t = 1$.

- 1) Statement I is True, Statement II is True
- 2) Statement I is False, Statement II is False
- 3) Statement I is True, Statement II is False
- 4) Statement I is False, Statement II is True

79. For the four circles M, N, O and P, following four equations are given;

Circle: $M : x^2 + y^2 = 1$

Circle: $N : x^2 + y^2 - 2x = 0$

Circle: $O : x^2 + y^2 - 2x - 2y + 1 = 0$

Circle: $P : x^2 + y^2 - 2y = 0$

If the centre of circle M is joined with centre of the circle N, further centre of circle N is joined with centre of the circle O, centre of circle O is joined with the centre of circle P and lastly, centre of circle P is joined with centre of circle M, then these lines form the sides of a:

- 1) Rhombus 2) Square 3) Rectangle 4) Parallelogram

80. Variable chords of the parabola $y^2 = 4ax$ subtend a right angle at the vertex. Then:

Statement I: locus of the feet of the perpendiculars from the vertex on these chords is a circle
Statement II: locus of the middle points of the chords is a parabola

- 1) Statement I is True, Statement II is True
- 2) Statement I is False, Statement II is False
- 3) Statement I is True, Statement II is False
- 4) Statement I is False, Statement II is True





(NUMERICAL VALUE TYPE)

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Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

81. If e_1 and e_2 are the eccentricities of the ellipse, $\frac{x^2}{18} + \frac{y^2}{4} = 1$ and the hyperbola, $\frac{x^2}{9} - \frac{y^2}{4} = 1$ respectively and (e_1, e_2) is a point on the ellipse, $15x^2 + 3y^2 = k$, then k is equal to
82. If the foot of the perpendicular from point $(4, 3, 8)$ on the line $L_1 : \frac{x-a}{l} = \frac{y-2}{3} = \frac{z-b}{4}, l \neq 0$ is $(3, 5, 7)$, then the shortest distance between the line L_1 and line $L_2 : \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$ is equal to $\frac{1}{\sqrt{k}}$, where $k = \underline{\hspace{2cm}}$
83. If the locus of the mid-point of the line segment from the point $(3, 2)$ to a point on the circle, $x^2 + y^2 = 1$ is a circle of radius r , then the length of diameter is equal to
84. For each positive integer n , consider the point P with abscissa n on the curve $y^2 - x^2 = 1$. If d_n represents the shortest distance from the point P to the line $y = x$ and $\lim_{n \rightarrow \infty} (n \cdot d_n)$ has the value equal to k . Then $16k^2 =$
85. A circle has the same centre as an ellipse and passes through the foci F_1 and F_2 of the ellipse, such that the two curves intersect in 4 points. Let P be any one of their point of intersection. If the major axis of the ellipse is 17 and the area of the triangle PF_1F_2 is 30, then the distance between the foci is
86. The maximum value of the function $f(x) = 2x^3 - 15x^2 + 36x - 48$ on the set $A = \left\{ x \mid x^2 + 20 \leq 9x \right\}$ is:





87. Find the number of integral values of parameter a for which three chords of the ellipse $\frac{x^2}{2a^2} + \frac{y^2}{a^2} = 1$ (other than its diameter) passing through the point $P\left(11a, -\frac{a^2}{4}\right)$ are bisected by the parabola $y^2 = 4ax$.
88. The area (in sq. units) of an equilateral triangle inscribed in the parabola $y^2 = 8x$, with one of its vertices on the vertex of this parabola, is $\lambda\sqrt{3}$, where $\lambda = \underline{\hspace{2cm}}$
89. Axis of a parabola lies along x-axis. If its vertex and focus are at distance 2 and 4 respectively from the origin, on the positive x-axis, then the point $(\alpha, -4)$ lies on the parabola, where $\alpha > 0$. Then the value of α is
90. The locus of the point of intersection of the lines $(\sqrt{3})kx + ky - 4\sqrt{3} = 0$ and $\sqrt{3}x - y - 4(\sqrt{3})k = 0$ is a conic, whose eccentricity is





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